This project, titled **Joint Crack Segmentation**, is an advanced computer vision pipeline designed to automate the identification and segmentation of structural anomalies, specifically focusing on **drywall joints** and **surface cracks**. It leverages state-of-the-art models like the Segment Anything Model (SAM) for data preparation and CLIPSeg for language-guided image segmentation.

Project Components & Objectives

The primary goal of the project is to create an "Enhanced Model" capable of precise pixel-level segmentation of cracks and construction joints. This is achieved through a multi-stage workflow:

1. **Label Refinement:** Converting coarse bounding boxes or polygons into high-fidelity masks using zero-shot foundation models.
2. **Hybrid Training:** Training a language-aware segmentation model that can identify specific areas based on text prompts.
3. **Accuracy Optimization:** Using specialized loss functions to handle fine-grained crack structures.

Data & Technologies Used

- **Datasets:** The model uses two distinct datasets sourced from Roboflow:
  - **Drywall Join Dataset:** Focuses on taping areas and drywall seams.
  - **Surface Crack Dataset:** Focuses on wall and structural cracks.
- **Key Frameworks:**
  - **Segment Anything (SAM):** Used to "perfect" training labels by generating masks from bounding boxes.
  - **CLIPSeg:** A transformer-based model used for promptable image segmentation.
  - **PyTorch & Transformers:** Core deep learning engine and model management.
  - **Albumentations:** For robust data augmentation (flips, rotations, brightness contrast).

Technical Methodology

1. Automated Mask Refinement (SAM)

The project uses the sam_vit_h (Vision Transformer Large) checkpoint to process raw labels. For datasets containing only bounding boxes (like the Drywall set), SAM automatically generates tight segmentation masks, which are then saved as the ground truth for the next phase.

2. CLIPSeg Model Fine-tuning

Instead of traditional class-based segmentation, the project utilizes **CLIPSeg**, which segments images based on text prompts.

- **Prompts:** The model is trained to recognize specific terms such as *"segment joint tape"* or *"segment wall crack"*.

- **Hybrid Loss:** The training utilizes a **Dice-BCE Loss**, which combines Binary Cross Entropy (pixel-wise accuracy) with Dice Loss (to better handle the overlap of thin, elongated structures like cracks).

3. Training Configuration

- **Epochs:** 25.
- **Optimizer:** AdamW with a learning rate of 2×10^{-5}.
- **Freeze Strategy:** The encoder is frozen during training to preserve pre-trained features and prevent overfitting on smaller structural datasets.

Results & Evaluation

The project includes a dedicated validation suite that calculates:

- **Mean Intersection over Union (mIoU):** To measure spatial overlap between predicted and actual cracks.
- **Dice Coefficient:** To evaluate the model's performance on the segmentation boundaries.
- **Visual Overlays:** The system generates side-by-side comparisons of the original image, the SAM-refined ground truth, and the model's final prediction heatmap.

How to Run

1. **Requirements:** Ensure all dependencies are installed via pip install segment-anything segmentation-models-pytorch transformers roboflow.
2. **Execution:** The core logic is contained within the VinayKamma Enhanced Model.ipynb notebook, which handles data downloading, refinement, training, and visualization in a single workflow.